Brain CT Hemorrhage Segmentation Pipeline Report

Week 5 — nnU-Net v2 training, augmentation audit and evaluation

1. Executive summary

This report documents the actual Week 5 nnU-Net run and validation evaluation. The configured model uses nine CT-derived channels and six output classes: background, IPH, SAH, EDH, SDH and IVH. Reported metrics are generated from the saved validation predictions; this is a research prototype and is not clinically validated.

2. Dataset and pipeline

Dataset: Dataset501_BrainCTICH25D. Configuration: 2d. Fold: 0. Training target: 50 epochs. Inputs: 9 windowed channels representing previous/center/next slices and brain/subdural/bone windows. Outputs: six class logits. Dataset files are reused rather than rebuilt.

3. Best checkpoint

Checkpoint: C:\Users\whate\nnUNet\nnUNet_results\Dataset501_BrainCTICH25D\nnUNetTrainerWeek5__nnUNetPlans__2d\fold_0\checkpoint_best.pth

Selection method: native nnU-Net best checkpoint (native selection metric; verify metric definition)

4. Validation summary

Validation cases: 33

Hemorrhage macro Dice (excluding background): 0.5399388528687288

Hemorrhage macro IoU (excluding background): 0.3907270669552599

5. Per-class validation metrics

Class	Support voxels	Dice	IoU	Precision	Recall
Background	293453711	0.9995	0.9990	0.9994	0.9996
IPH	363848	0.7836	0.6441	0.7689	0.7988
SAH	78139	0.3691	0.2264	0.4084	0.3368
EDH	39920	0.6367	0.4670	0.6419	0.6316
SDH	46206	0.2986	0.1755	0.3136	0.2849
IVH	143744	0.6117	0.4406	0.7548	0.5143

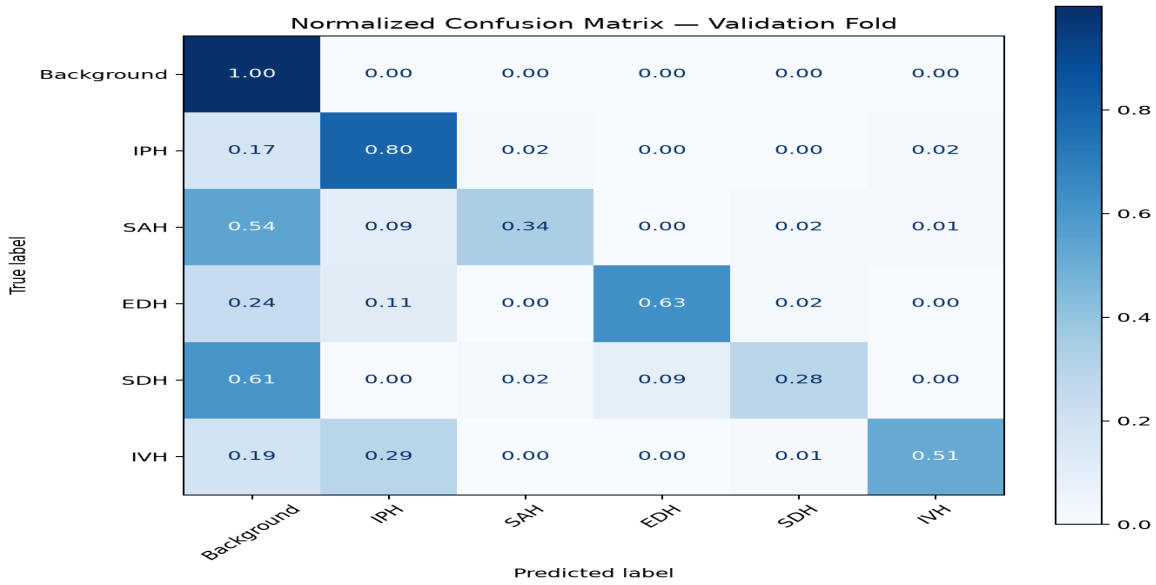
6. Training history and plots

Training curves are included only when the corresponding native nnU-Net logs or checkpoint logger history were available. See the report output folder for CSV history and PNG plots.

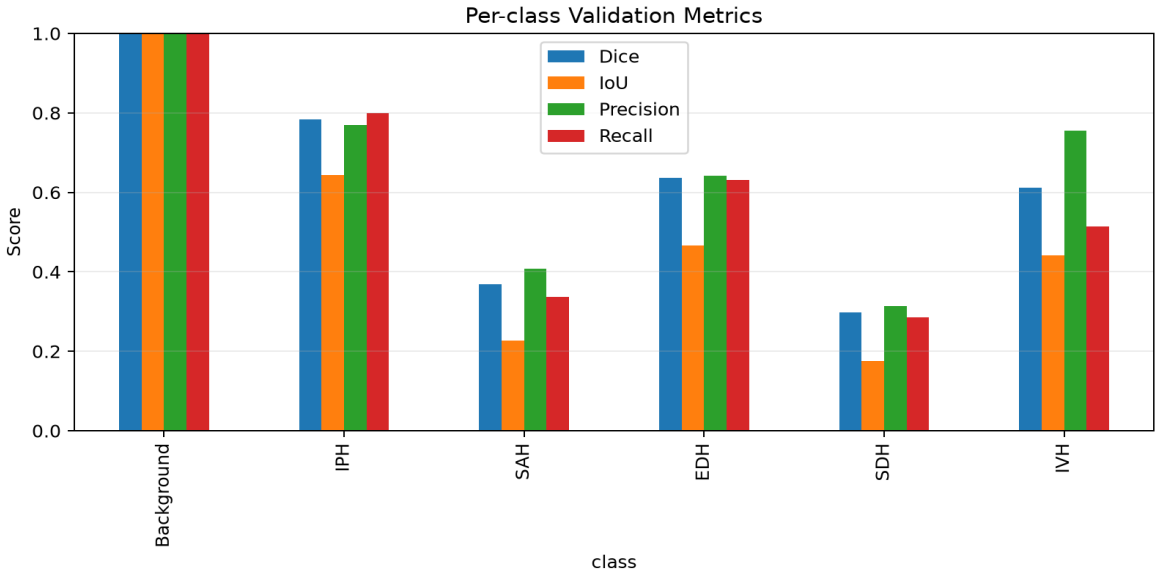
Native nnU-Net training progress



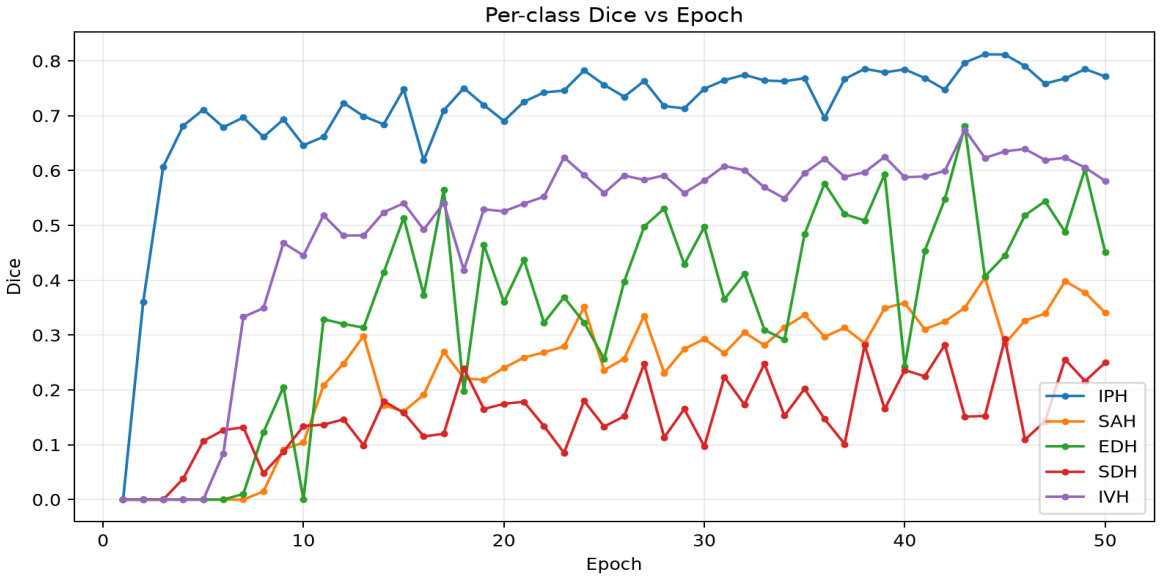
Normalized validation confusion matrix



Per-class validation metrics



Per-class Dice vs epoch



7. Augmentation compliance and limitations

See augmentation_audit.json for the source-level checks against the installed nnU-Net transform implementation. This dataset stores nine windowed CT channels, not raw HU volumes. Therefore, true brightness/contrast jitter bounded to ±50 HU is not claimed as implemented. If the Week 5 specification requires exact HU-space augmentation, the pipeline must apply it to raw-HU images before windowing. Native transform source inspection is not by itself proof that every transform was active during training.

8. Limitations

Metrics are computed on the selected nnU-Net fold validation cases and are not independent external validation. A class with zero Dice should be reported as a model limitation, not hidden. The model is not approved for clinical diagnosis or treatment decisions.